EC 571: Digital VLSI Circuit Design

Gate Delay and Logical Effort

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Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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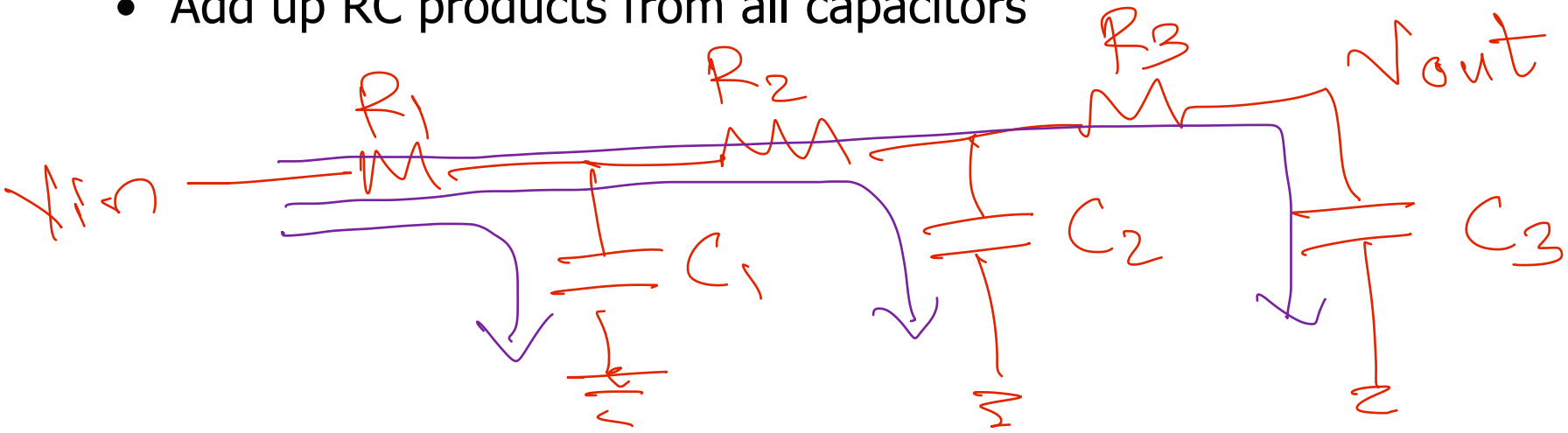
Last lecture

- Last lecture
  - Inverter delay optimization
- Today's lecture
  - Gate delay and Logical effort
- Reading
  - Chapter 6

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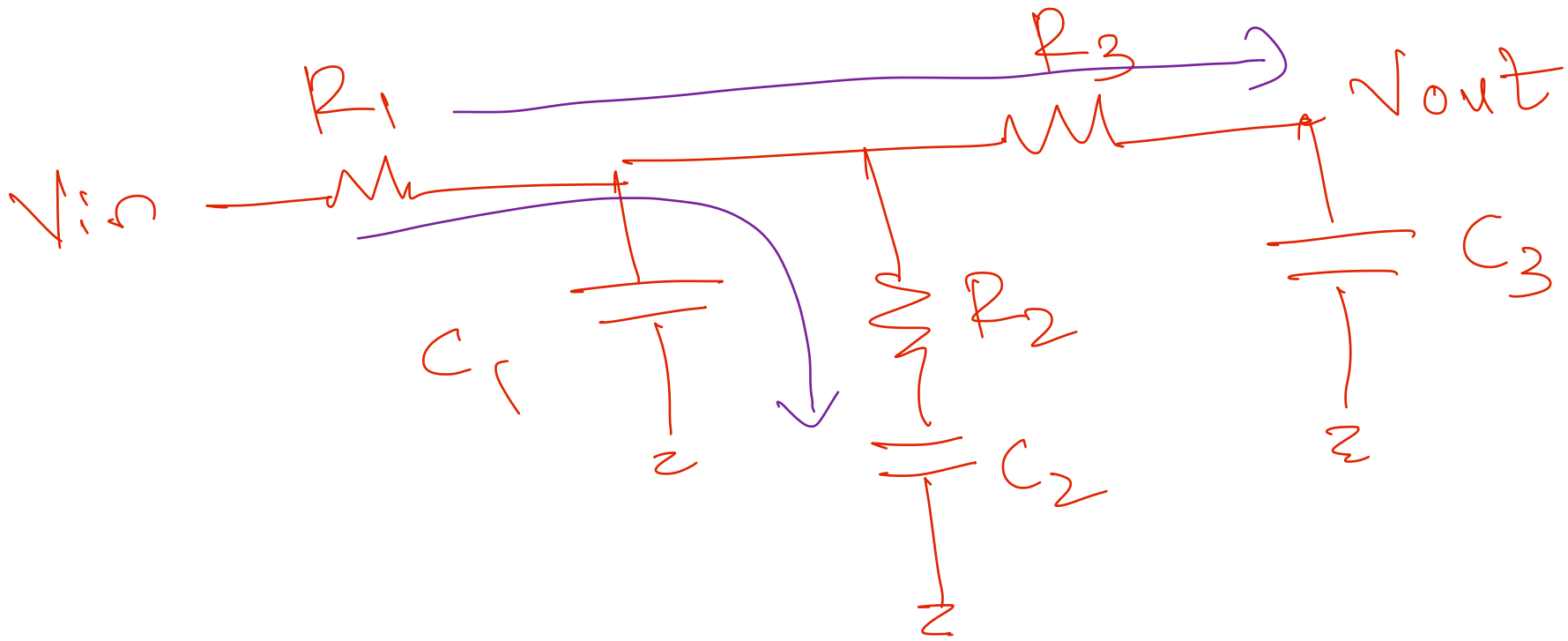
Elmore Delay

- Steps to calculate Elmore RC delay
  - For each capacitor draw path of current from capacitor to input
  - Multiply each capacitance value with the sum of resistances on the current path that are common with the path from Vin to Vout
  - Add up RC products from all capacitors



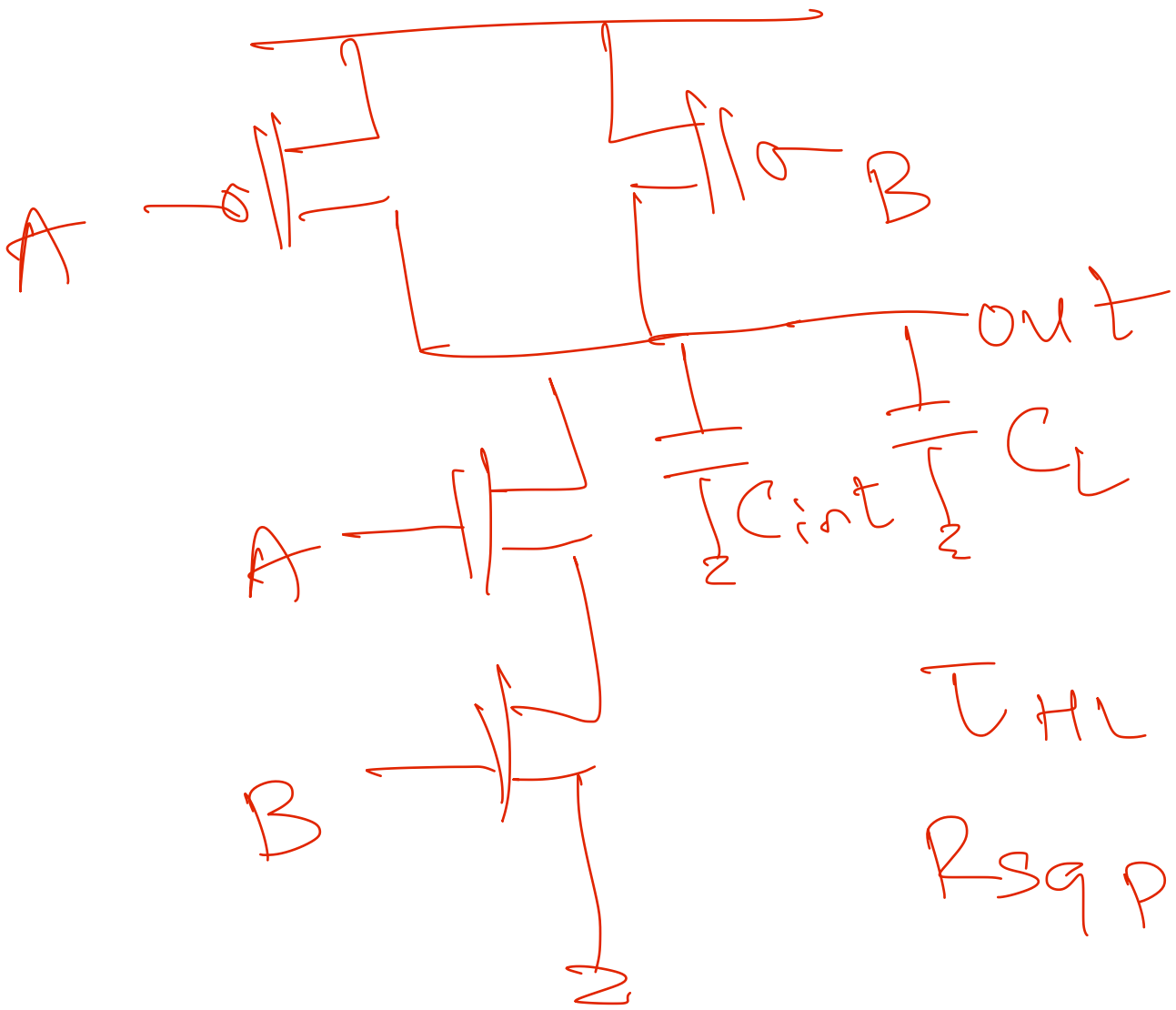
$$T = R_1C_1 + (R_1+R_2)C_2 + (R_1+R_2+R_3)C_3$$

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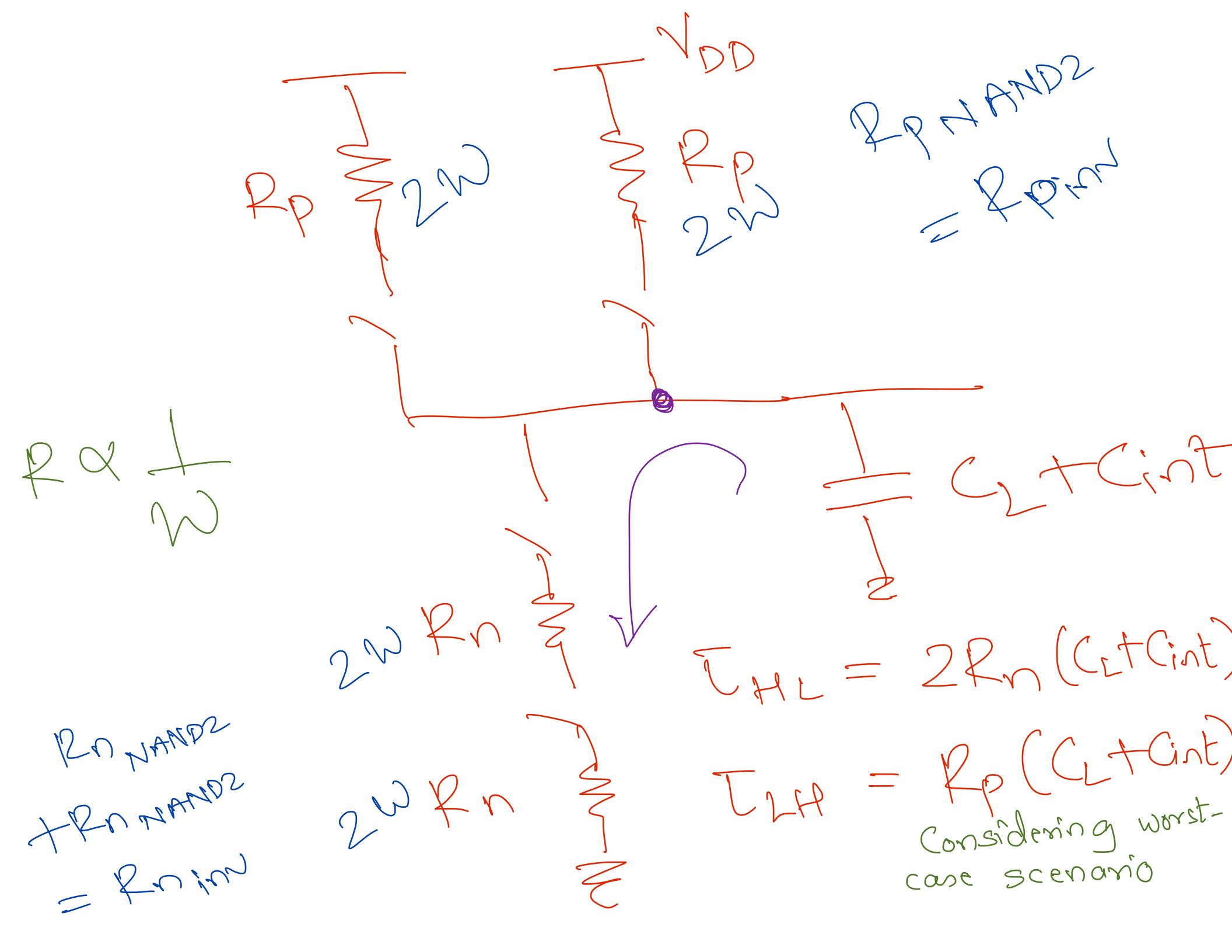
$$\tau = R_1 C_1 + R_1 C_2 + (R_1 + R_3) C_3$$

NAND 2



$$\tau_{HL} = \tau_{LH}$$

$$R_{sqp} = 2 R_{sqN}$$


V_{DD}
R_p
2W
R_p
2W
$R_{p\,NAND2}$
$= R_{p\,inv}$
$R \propto \frac{1}{W}$
2W R_n
2W R_n
$C_L + C_{int}$
$\tau_{HL} = 2R_n(C_L + C_{int})$
$\tau_{LH} = R_p(C_L + C_{int})$
Considering worst-case scenario
$R_{n\,NAND2}$
$+R_{n\,NAND2}$
$= R_{n\,inv}$

$$C_{in_{NAND2}} = C_G(W_p + W_n)$$

$$= C_G(2W + 2W)$$

$$= 4W C_G$$

$$C_{int_{NAND2}} = C_D(W_p + W_p + W_n)$$

$$= C_D(2W + 2W + 2W)$$

$$= 6W C_D$$

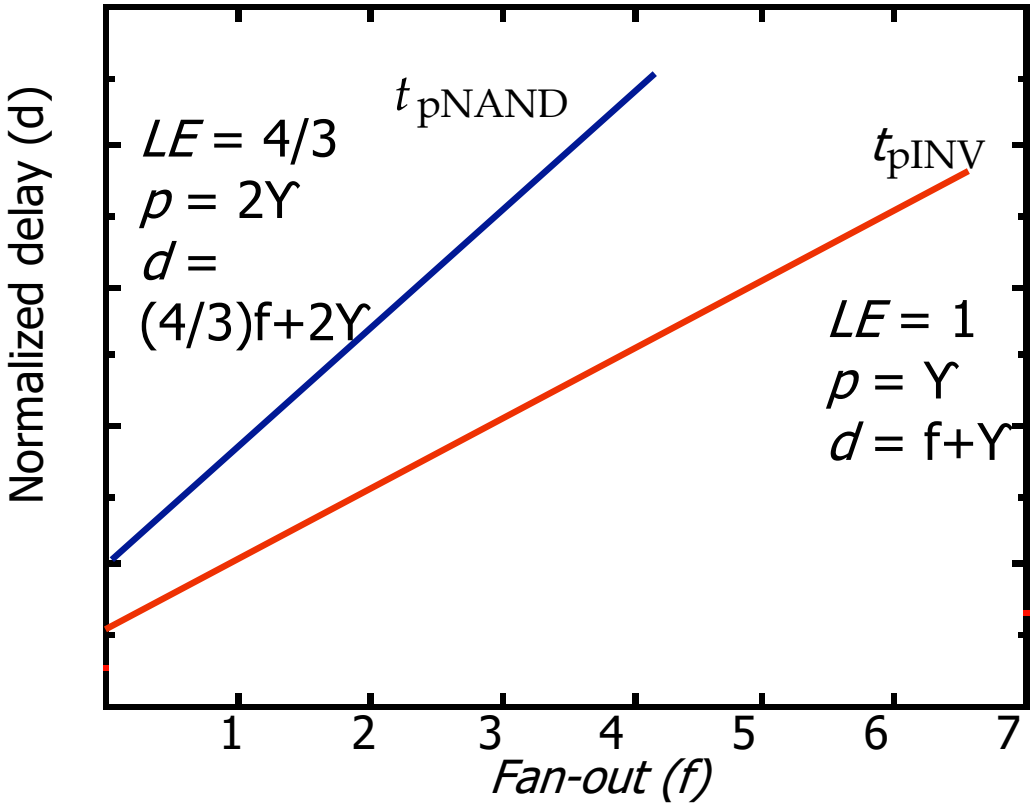
Logical Effort - Catalog of gates

Gate type	Logical Effort for different number of inputs					
	1	**2**	**3**	**4**	**5**	**n**
Inv	1					
NAND		4/3	5/3	6/3	7/3	(n+2)/3
NOR		5/3	7/3	9/3	11/3	(2n+1)/3
Mux		2	2	2	2	2
XOR,XNOR		2	6	16		

Gate type	Parasitic delay
Inv	$p_{inv} = 1$
n-input NAND	$n.p_{inv}$
n-input NOR	$n.p_{inv}$
n-way Mux	$2n.p_{inv}$
n-input XOR,XNOR	$4n.p_{inv}$

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Logical Effort - INV vs NAND



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Method of Logical Effort

- Logical effort of gate i = LE_i
- Stage effort: $h_i = LE_i f_i$
- Path electrical effort: $F = C_{out}/C_{in}$
- Path logical effort: $G = LE_1 LE_2 \ldots LE_N$
- Branching effort: $B = b_1 b_2 \ldots b_N$
- Path effort: $H = GFB$
- Path delay $D = \Sigma d_i = \Sigma p_i + \Sigma h_i$

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Method of Logical Effort

- Compute the path effort: H = GBF
- Find the best number of stages: N ~ $\log_4 H$
- Compute the stage effort: h = $H^{1/N}$
- Sketch the path with this number of stages
- Work from either end, find sizes:
  $C_{in} = C_{out} * LE/h$

Reference: Sutherland, Sproull, Harris, "Logical Effort, Morgan-Kaufmann 1999.

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